

Education & Communication

Ignite curiosity, enlighten minds.

Nothing in life is to be feared, it is only to be understood. Now is the time to understand more, so that we may fear less. —Marie Curie

Cancer related

Information Assessment and Recommendations

When we conduct cancer-related searches on different platforms, we often come across some contradictory opinions and emotional expressions. Meanwhile, "online medical consultation" has always been a phenomenon that occurs frequently while being criticized for its inaccuracy. With the popularization of AI technology, many patients and the general public may turn to AI assistants for consultation. Although the object of consultation may change, the accuracy and the impact of technology on people's decision-making remain lingering issues. For many people, the word "cancer" is terrifying. When this fear is not regulated by rational scientific perception, it is likely to exacerbate the psychological burden on patients or become an incentive for some inappropriate intervention.

In the cancer-related science popularization section, we hope to focus on evaluating whether people's understanding of cancer is correct in the context of the current era with information overload and the AI trend, and how we can use online media and AI tools to better promote relevant knowledge, enabling people to understand cancer properly and deal with it appropriately.

- **Survey**

General public:

Questionnaire

Q :

-basic information like age and education level

-what knowledge do you have about cancer (e.g. Causes, risk factors, current treatment, popularity, symptoms,etc.)

- some scenarios with common misunderstandings to test if they hold the same opinion
- where did you learn it from (experts, videos, search engines, AI,etc.)
- rank these information resources from what you think is the most reliable to the least you trust.
- how do you feel about cancer ?
- what's your frequency of conducting a body check ?
- do you smoke? If you do, how do you perceive the connection between smoking and developing lung cancer?
- What do you feel about current public education on cancer ? How would you like it to be improved?

Patients:

Interview (along with one in iHP

Q :

- The things you wish to know earlier about cancer.
- Your comment on current treatment.

Medical workers:

Interview

Q:

- have you encountered cases where people blindly trust information from unreliable sources at work?
- What are the content and reasons for such misunderstandings?
- What challenges would be brought from those misunderstandings?
- What basic information is currently urgently needed for the public to understand?
- If one wants to search for relevant information, which resources are recommended?
- What are the strategies people could employ to reduce the risk of developing cancer?
- Is it possible for us to do science popularization for cancer patients?

Public health-related institutions and professional science popularization workers:

Interviews

Q:

- their understanding of the issues
- their experience in science popularization.

- **Target audience:** public, patients
- **Content:**
 1. **Evaluation Report:** Based on the results of the survey, we intended to form a report regarding the current situation of people's understanding of cancer through different media and the reliability of different media;
 2. **handbook:** A promotional handbook that, based on the opinions of experts in fields such as medical workers, guides people to better access cancer-related information, manage symptoms, and view disease diagnoses;
 3. **science popularization:** Organize the thoughts and publish content on major online platforms, then hold offline publicity activities targeting the public
 4. **Community:** Share the content of the handbook and some experiences gained from previous held activities with workers engaged in similar work in the community, and explore the possibility of long-term cooperation.
 5. **Hospital visits:** Have conversations with cancer patients to understand their needs and provide them with useful information and mental support.

Cancer Prevention Popularization

"Prevention is better than cure."

Many cancers can be avoided through early prevention. Take small cell lung cancer as an example. The occurrence of this cancer is highly correlated with smoking. Prevention is the best treatment. By popularizing the causes of cancer and advocating a healthier lifestyle, we hope to promote public health .

- **Target audience:** public, students
- **Content:** cause & prevention, symptoms & diagnosis of cancer (the impact of

smoking...

The promotional methods can be carried out together with the previous event, while adding some science popularization activities on campus .

To ensure that the content is easy to understand, we can explore forms of expression such as comics/scripts around the theme and design science popularization activities that can better encourage people to participate.

We can also develop online courses or short promotion videos introducing the current treatments. Providing information including principles, as well as the effects, costs, possible side effects, precautions, etc. of these therapies.

Death Education

- **Target audience:** children; patients and their caregivers
- **Content**

1. Understand the topic:

Death is a heavy topic that many people choose to avoid for a long time, yet it is something everyone will eventually face. Meanwhile, death education is also a form of "education for life". Only by understanding death can people appreciate the value of life.

Precisely because it is an important issue, we hope to provide death education to children, patients, and their families to help them better face it. And because it is a complex issue, we cannot rely solely on our own knowledge. Instead, we will communicate with patients, consult experts in related fields, and carefully design how to deliver the content when presenting this topic.

It's a rather delicate topic for cancer patients, so we would consult experts like psychotherapist before and have conversations with the caregivers rather than directly with the patients and ask for their approval.

2. Children's book: we can form an illustration to show what death is and that life and death are both components of life.

3. Pamphlets: we would first do some research on topics like palliative care and consult with the hospital. We will make pamphlets with useful information and death education content. We will then check with the hospital to see if these can be handed out to patients .

Synthetic biology related

- Survey:

Questionnaires

about knowledge , acceptance of related products, etc.

- Target audience: public, students
- Content:

1. Handbook & shared teaching materials:

Make handbooks in synthetic biology education (some examples form previous teams can serve as valuable resource), while designing popular science content and forms based on actual situations, collaborating with professional educators from schools and other institutions, attempting to launch similar science brochures or materials containing teaching materials such as courseware. We can first conduct trial lectures in classrooms and obtain feedback. Then integrate them into teaching sessions to become part of the teaching system itself;

2. **Videos** :The content sorted out above can be made into forms such as videos and published on online platforms;

3. **Publicity**: targeting the general public , guided by application examples to familiarize people with this new technology and its derived industries;

4. Website & network:

Many previous projects have launched synthetic biology resources such as illustrations and handbooks for students at different levels. We can try to connect with these teams, integrate these resources, add our own work, and help promote them so that good teaching resources can be accessed by more people.

If there are partner teams from other countries, with the consent of the relevant teams, we can also carry out translation work on the materials.

We can cooperate to complete more systematic curriculum design and record online courses. (e.g. On nobel prize, our school has organization which carry out similar activities, we can also cooperate with them.)

If we have the ability, we can try to cooperate in building a large language model embedded in the webpage to answer questions related to synthetic biology;

5. Meeting :

Several iGEM teams can jointly organize a meeting (online to cover as many regions as possible, offline for teams within Hangzhou), with the theme of sharing project design and synthetic biology education, targeting university students. Different teams can both provide mutual evaluation and suggestions and are responsible for promoting the event on their own campuses to recruit audiences. In this way, the same event can not only obtain important feedback on projects but also achieve educational goals, and co-organizing can attract a larger and more diverse audience, making the opinions more comprehensive and diverse.